TÉCNICO
UNIVERSIDADE
DE LISBOA

0.1 · COURSE PRESENTATION

# Ship and Ocean Systems Design

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COURSE OVERVIEW

# Objectives

## General

- Concept design
- Design process
- Laws and regulations
- Types of ships

## Operational

- Main characteristics of the ship
- Optimization methods
- Hull form design
- Calculations and Specifications
- Financial analysis

# Syllabus

1 **Design Process**
Design stages · Design spiral · Sequential and concurrent engineering · Actors · Design documents · Ship Product Data Model

2 **Ship Model – Initial Dimensioning**
Process of initial dimensioning · Estimation methods

3 **Financial Model**
Voyage model: cost and time · Newbuilding cost · Shipbuilding contract and key annexes · Ship financing · Operating costs · Investment analysis

4 **Synthesis Model and Optimization in Ship Design**
Synthesis model · Single- and multi-objective optimization

5 **Design Criteria: Statutory Regulations and Classification Rules**
IMO: Load Lines, SOLAS, MARPOL (EEDI, CII), COLREG, Tonnage · EU: ETS, FuelEU Maritime, Directives · ILO, OCIMF · National Laws

6 **Main Types of Ships**
Classification · Main types · Purpose and features

7 **Hull Form, Lines Plan and General Arrangement**
Displacement distribution · Sections, bow, stern · Lines plan · GA

LEARNING OUTCOMES

# After the Course, You Can…

| | | |
|---|---|---|
| 1 | **Frame a design problem** | Turn the owner's requirements (cargo, capacity, route, ports, speed, fuel, rules) into a design basis |
| 2 | **Place the ship in the design process** | Name the design stages, actors and contract documents; classify the ship by its dominant constraint |
| 3 | **Estimate the main characteristics** | L, B, D, T, CB, displacement, weights, power and capacities from regressions and methods |
| 4 | **Value the design** | Build a voyage model and a cost model: building cost, financing, operating cost |
| 5 | **Validate and check compliance** | Compare with reference ships; check freeboard, stability and draught at every port; energy efficiency and emissions: EEDI, CII, EU ETS, FuelEU Maritime |
| 6 | **Optimize** | State variables, constraints and objectives; solve single- and multi-objective problems |
| 7 | **Sketch the hull form and GA** | Turn the optimized dimensions into a hull form and a general arrangement, with a hydrostatics check |
| 8 | **Recognize ship types** | Describe the systems and equipment of the main merchant ship types and the rules behind them (not tested) |
| 9 | **Specialize and defend** | Carry two specializations to a calculation note; present and justify the design in 15 minutes |

# Evaluation Method

| Component | Description | Format | Weight |
|---|---|---|---|
| Evaluation Test | Design process; Law and Regulations reading | Multiple-choice | 20% |
| Ship Design Project | Concept design of a merchant ship. Synthesis model and optimization. | Report | 60% |
| Project Presentation | Presentation, discussion and justification of the Project. | Presentation | 20% |

**Attendance and exercises:** attendance is not compulsory but highly recommended. In-class exercises are individual practice on a reference ship, not graded; the project is group work done outside class.

**Weekly self-assessment quizzes (ungraded).** They show where you stand against the learning objectives and how deep the course goes.

**Programming skills and software proficiency:** 3D modelling, 2D drafting, programming and/or specific software use are neither taught nor evaluated in the present course. If you lack these skills, please consult the bibliography to reach a basic level of proficiency.

**Academic integrity and use of AI.** All submitted work must be your group's own. Copying from other groups, past reports or uncredited sources is plagiarism and will not be accepted. AI tools may be used as support, provided their use is declared in the report. You must be able to explain, justify and reproduce everything you submit; this is checked in the project presentation.

# Lecture Calendar

## Weekly timetable

| Day | Time | Room |
|---|---|---|
| Wednesday | 13:00–15:00 | E2 |
| Wednesday | 15:00–16:30 | E2 |
| Friday | 13:30–15:30 | E2 |
| Friday | 15:30–17:00 | E1 |

28 sessions over 7 weeks, from 4 November to 18 December.

## Dates and milestones

| Week | Date | Milestone |
|---|---|---|
| 1 | 6 Nov | Project groups formed |
| 2 | 13 Nov | Ship Design Project assigned |
| 4 | 27 Nov | Evaluation Test |
| 6 | 13 Dec | Ship Design Project due |
| 7 | 18 Dec | Project Presentation |

Interested in a particular ship type? Present your arguments; the lecturer will consider them.

## Class Structure

- "Types of Ship" runs as a continuous theme throughout the calendar.
- Each lecture session ends with an individual exercise (not a group exercise) on the reference ship (not graded).

# Office Hours

## Sessions to discuss project

Book at calendly.com/vnsd-luisbatista/meetme

- 2 hours per week, in 30-minute slots, one group per slot.
- Each group may book no more than one slot per week.
- Upon scheduling, specify the purpose of the meeting and clearly state the points to be discussed.
- Location: Vera Navis Ship Design (VNSD), Av. 5 de Outubro nº 72 9º A

# Course Plan · 28 Sessions

| Week | Wed 13:00–15:00 | Wed 15:00–16:30 | Fri 13:30–15:30 | Fri 15:30–17:00 |
|---|---|---|---|---|
| **Week 1**<br>4–6 Nov | **W1.1**<br>**Course + project** | **W1.2**<br>Requirements + process | **W1.3**<br>Main dimensions | **W1.4**<br>Displacement equation |
| **Week 2**<br>11–13 Nov | **W2.1**<br>Human factors 1/2 | **W2.2**<br>Human factors 2/2 | **W2.3**<br>Dimensions · Rules | **W2.4**<br>Power estimate |
| **Week 3**<br>18–20 Nov | **W3.1**<br>Weights · deadweight | **W3.2**<br>Hold capacity | **W3.3**<br>Machinery · propeller | **W3.4**<br>Voyage model |
| **Week 4**<br>25–27 Nov | **W4.1**<br>Building cost | **W4.2**<br>Financing · RFR | **W4.3**<br>Stability · EEDI | **W4.4**<br>**Evaluation Test** |
| **Week 5**<br>2–4 Dec | **W5.1**<br>Synthesis model | **W5.2**<br>Single-objective | **W5.3**<br>Multi-objective | **W5.4**<br>Hull form 1 |
| **Week 6**<br>9–11 Dec | **W6.1**<br>Hull form 2 | **W6.2**<br>Lines plan · GA | **W6.3**<br>**Project clinic 1/2** | **W6.4**<br>**Project clinic 2/2** |
| **Week 7**<br>16–18 Dec | **W7.1**<br>Digital transition | **W7.2**<br>AI in ship design | **W7.3**<br>**Presentations 1/2** | **W7.4**<br>**Presentations 2/2** |

Design process · Ship model · Voyage and cost · Optimization and digital · Hull form and GA · Guest lecturer

Fixed: course, test, clinics, presentations

COURSE MATERIALS · GITHUB

# Everything for the Course in One Place

**Course site: open it, bookmark it**

**l-u-i-s-b-a-t-i-s-t-a.github.io/PNSO-2026**

Repository: github.com/L-u-i-s-B-a-t-i-s-t-a/PNSO-2026

## What is inside

**The course site:** the seven weeks, session by session, with each session's materials beside it

**week1 to week7:** slides, readings and exercise answers as PDFs, one folder per week

**sources:** open data and reports used in the course

## How to use it

Materials appear about a week before each session; dashed items are not out yet. Exercise answers follow after each week. Always work from the site, not from old copies.

# Week 1 · 4–6 November

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W1.1**<br>**Wed 4 Nov**<br>**13:00–15:00** | **Course presentation + Ship Design Project**<br>Classification of Ship Types | 0.1 and 0.2 [2]; Reading 1.1 | Groups and project questions<br>**Activity 1.1** Find ships in the StatCode 5 Explorer (15 min) | — |
| **W1.2**<br>**Wed 4 Nov**<br>**15:00–16:30** | **Owner's requirements; design process: phases, design spiral**<br>Deadweight, volume and linear-dimension carriers | §1.3.1–1.3.5, §1.3.6.1–1.3.6.2 [1]; 0.6 [2] | **Exercise 1.2** Does the owner's brief add up? Classify<br>**Exercise 1.3** Which design phase made this document?<br>**Activity 1.3** The ship's documents | **D1** |
| **W1.3**<br>**Fri 6 Nov**<br>**13:30–15:30** | **Design methods; displacement estimate; main dimensions**<br>Container ships | §1.3.6–1.3.7; §2.1–2.2; App. A [1] | **Exercise 1.4** First Δ, L, B, T, D from the App. A regressions, with 95% bands<br>Homework H0 (optional) | **D2** |
| **W1.4**<br>**Fri 6 Nov**<br>**15:30–17:00** | **Displacement equation**<br>Bulk carriers | §2.13; App. C (Normand) [1] | **Exercise 1.5** Displacement equation exercise, Q1–Q6<br>Homework H1 (optional) | **D3** |

**Milestone:** project groups formed, Fri 6 Nov.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.
[2] 0.1 Course Presentation, 0.2 Ship Design Project and 0.6 The Reference Ship (course documents).

# Week 2 · 11–13 November

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W2.1**<br>**Wed 11 Nov**<br>**13:00–15:00** | **Human factors in ship design (1 of 2)**<br>Guest lecturer | Guest lecturer's material | Set by the guest lecturer | — |
| **W2.2**<br>**Wed 11 Nov**<br>**15:00–16:30** | **Human factors in ship design (2 of 2)**<br>Guest lecturer | Guest lecturer's material | Set by the guest lecturer | — |
| **W2.3**<br>**Fri 13 Nov**<br>**13:30–15:30** | **Length, beam, depth, draft, form coefficients · Law and Regulations introduction (20 min)**<br>General cargo, multi-purpose and heavy-lift ships, reefers | §2.3–2.12; §2.7.1 [1] | **Exercise 2.3** Check L, B, T, D, CB and Fn, L/B, B/T, L/∇^(1/3) against the book's ranges; redo at 15 kn | **D3** |
| **W2.4**<br>**Fri 13 Nov**<br>**15:30–17:00** | **Preliminary power estimate**<br>Ro-Ro ships and vehicle carriers | §2.14; §5.1 [1] | **Exercise 2.4** Power by Admiralty and Holtrop–Mennen; compare | **D3** |

**Law and Regulations reading** starts on 13 Nov, for the Evaluation Test on 27 Nov.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

# Week 3 · 18–20 November

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W3.1**<br>**Wed 18 Nov**<br>**13:00–15:00** | **Weights and centroids; deadweight; displacement check**<br>Crude oil and shuttle tankers | §2.15–2.16 [1] | **Exercise 3.1** Lightship by two methods; close the weight balance | **D3** |
| **W3.2**<br>**Wed 18 Nov**<br>**15:00–16:30** | **Hold capacity**<br>Product and chemical tankers | §2.17 [1] | **Exercise 3.2** Hold volume against the target | **D3, D8** |
| **W3.3**<br>**Fri 20 Nov**<br>**13:30–15:30** | **Main machinery and propeller**<br>Gas carriers: LNG, LPG, ammonia, CO2 | §5.2–5.3 [1] | **Exercise 3.3** Engine selection and propeller diameter<br>Homework H2 (optional) | **D3** |
| **W3.4**<br>**Fri 20 Nov**<br>**15:30–17:00** | **Voyage model**<br>Ro-Pax and fast ferries | §1.2.1 [1] · Voyage notes A [2] | **Exercise 3.4** Round-trip time and fuel on the reference route | **D4** |

> **Milestone:** ship model complete (D3).

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] Voyage and Finance Notes – Part A: The Voyage Model (course notes).

# Week 4 · 25–27 November

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W4.1**<br>**Wed 25 Nov**<br>**13:00–15:00** | **Building cost**<br>Cruise ships and yachts | §6.2–6.5 [1] ·<br>Shipbuilding contract [2] | **Exercise 4.1** Newbuilding cost by weight group | **D4** |
| **W4.2**<br>**Wed 25 Nov**<br>**15:00–16:30** | **Financing, operating costs, required freight rate**<br>Offshore oil and gas 1: drilling units | Finance notes B [3] ·<br>How to finance a ship [4] | **Exercise 4.2** RFR and NPV; sensitivity to speed<br>Homework H3 (optional) | **D4** |
| **W4.3**<br>**Fri 27 Nov**<br>**13:30–15:30** | **Stability, trim, freeboard; EEDI**<br>Offshore oil and gas 2: production and storage units | §2.18–2.19; §1.2.2 [1] | **Exercise 4.3** GM, freeboard and attained EEDI<br>Homework H4 (optional) | **D7** |
| **W4.4**<br>**Fri 27 Nov**<br>**15:30–17:00** | **Evaluation Test (20%)** | Law and Regulations reading [5] | Multiple choice | **Test** |

**Evaluation Test, Fri 27 Nov:** design process and the Law and Regulations reading. Not tested: stability and freeboard, ship types, human factors.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.
[2] Reading 4.1, Shipbuilding Contract · [3] Voyage and Finance Notes, Part B · [4] Reading 4.2, How to Finance a Ship · [5] Law and Regulations handout

# Week 5 · 2–4 December

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W5.1**<br>**Wed 2 Dec**<br>**13:00–15:00** | **The ship design optimization problem; ship synthesis model**<br>Offshore oil and gas 3: support vessels | §6.1; App. F [1] | **Exercise 5.1** Link ship, voyage and cost models; tornado chart | **D6** |
| **W5.2**<br>**Wed 2 Dec**<br>**15:00–16:30** | **Single-objective optimization**<br>Offshore oil and gas 4: offshore construction | §6.1; App. F [1] | **Exercise 5.2** Optimization: lecturer demonstration, then an individual step<br>Demo 5 · synthesis model | **D6** |
| **W5.3**<br>**Fri 4 Dec**<br>**13:30–15:30** | **Multi-objective optimization and MADM**<br>Offshore wind 1: installation vessels | App. F (AFRAMAX case) [1]; TOPSIS [2] | **Exercise 5.3** Pareto front RFR vs EEDI; pick one design with TOPSIS | **D6** |
| **W5.4**<br>**Fri 4 Dec**<br>**15:30–17:00** | **Hull form: distribution of displacement, waterlines**<br>Offshore wind 2: foundations, cable-layers | §3.1–3.2 [1] | **Exercise 5.4** Sectional area curve and LCB<br>Homework H5 (optional) | **D8** |

**Specialization checkpoint** in office hours this week.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] Hwang, C. L. and Yoon, K. (1981). *Multiple Attribute Decision Making*. Springer.

# Week 6 · 9–11 December

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W6.1**<br>**Wed 9 Dec**<br>**13:00–15:00** | **Hull form: sections, bow and bulbous bow, stern; hull series**<br>Offshore wind 3: service vessels (SOV, CTV) | §3.3–3.5; App. B [1] | **Exercise 6.1** Sections, bow and stern; bulb or not (§3.4.2); parent hull from App. B | **D8** |
| **W6.2**<br>**Wed 9 Dec**<br>**15:00–16:30** | **Lines plan, general arrangement, capacity plan**<br>Naval 1: surface combatants | §4.1–4.6; §2.7.1 [1] | **Exercise 6.2** GA sketch: collision bulkhead, engine room, double bottom; capacity plan | **D8** |
| **W6.3**<br>**Fri 11 Dec**<br>**13:30–15:30** | **Project clinic (1 of 2)** | Report and datasheet templates [2] | Review of each group's model and report | **Project** |
| **W6.4**<br>**Fri 11 Dec**<br>**15:30–17:00** | **Project clinic (2 of 2)** | Report and datasheet templates [2] | Review of each group's model and report | **Project** |

> **Ship Design Project due Sun 13 Dec (60%):** report and files, one ZIP by e-mail.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

[2] 0.3 Report Template and 0.4 Datasheet Template (course documents).

# Week 7 · 16–18 December

| Session | Topic · ship types (10 min) | Reading | Individual exercise | Practice for |
|---|---|---|---|---|
| **W7.1**<br>**Wed 16 Dec**<br>**13:00–15:00** | **Ship design in the digital transition**<br>Naval 2: patrol, coastguard and mine countermeasure vessels, uncrewed vessels | App. G §G.2–G.6; App. F [1] | **Exercise 7.1** Sobol variants of L, B, D, CB through the synthesis model; RFR against EEDI (demonstration + individual step) | **D6** |
| **W7.2**<br>**Wed 16 Dec**<br>**15:00–16:30** | **Artificial intelligence in ship design; AI demonstration**<br>Naval 3: amphibious and support ships, submarines | App. G §G.7–G.8 [1] | **Exercise 7.2** Surrogate model trained on Exercise 7.1's variants; re-optimize and check with the full model<br>Demo 7 · surrogate model | **D6** |
| **W7.3**<br>**Fri 18 Dec**<br>**13:30–15:30** | **Project presentations (1 of 2)** | — | 15-min group presentation, then discussion | **Pres.** |
| **W7.4**<br>**Fri 18 Dec**<br>**15:30–17:00** | **Project presentations (2 of 2)** | — | 15-min group presentation, then discussion | **Pres.** |

**Project Presentation, Fri 18 Dec (20%):** presentation, discussion and justification of the project, about 25 min per group.

[1] Papanikolaou, A. (2026). *Ship Design: Methodologies of Preliminary Design*, 2nd ed. Springer.

# Bibliography


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